



Sri Lanka Institute of Information Technology



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Greenhouse Automated System.
Proposal Document
P5-17

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BACKGROUND

Greenhouses do an important job of stabilizing the growing environment by modulating the ambient temperature and protecting the plants from outside harm. Manufacturers and farmers struggle with keep the conditions optimal within greenhouses. The problems include:

- 1) Water management- Plants need right water amount at the right time. Traditional methods are failed to manage the water levels accurately.
- 2) Balanced Temperature-Plants always need a proper temperature for germination. Varying control of temperature can cause damage to plants.
- 3) Insect control- Insects also become a huge problem in greenhouses. So, protecting from insects is necessary to prevent damage to plants.

To solve these issues, we suggest the architecture and develop of a smart greenhouse monitoring system. In this system we're going to control the soil moisture, temperature, Insect activity and watering the plants.

PROBLEMS AND MOTIVATION

Problems:

In the Greenhouse concept, providing water at the right time, feeding the plant at the right temperature and protecting the plant from insects are mandatory components to maintain the quality of the plants.

But to consumers,

- ☐ Not being able to see how much water is in the plant.
- ☐ Inability to provide water when needed.
- ☐ Inability to provide water from any place.
- ☐ Failure to ensure that the plant is at the proper temperature.
- ☐ Not being able to check if insects visit the plant.

Motivation:

As our solution Using a mobile phone,

- ☐ Being able to take care of the percentage of water in the plants.

- Being able to automatically water plants from anywhere.
- Being able to check if the plants are at the right temperature.
- Being able to check if insects visit the plants

AIM AND OBJECTIVE

Aim:

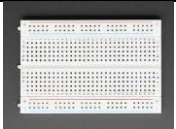
- To introduce a productive and convenient system to keep the plants in the greenhouse healthy.

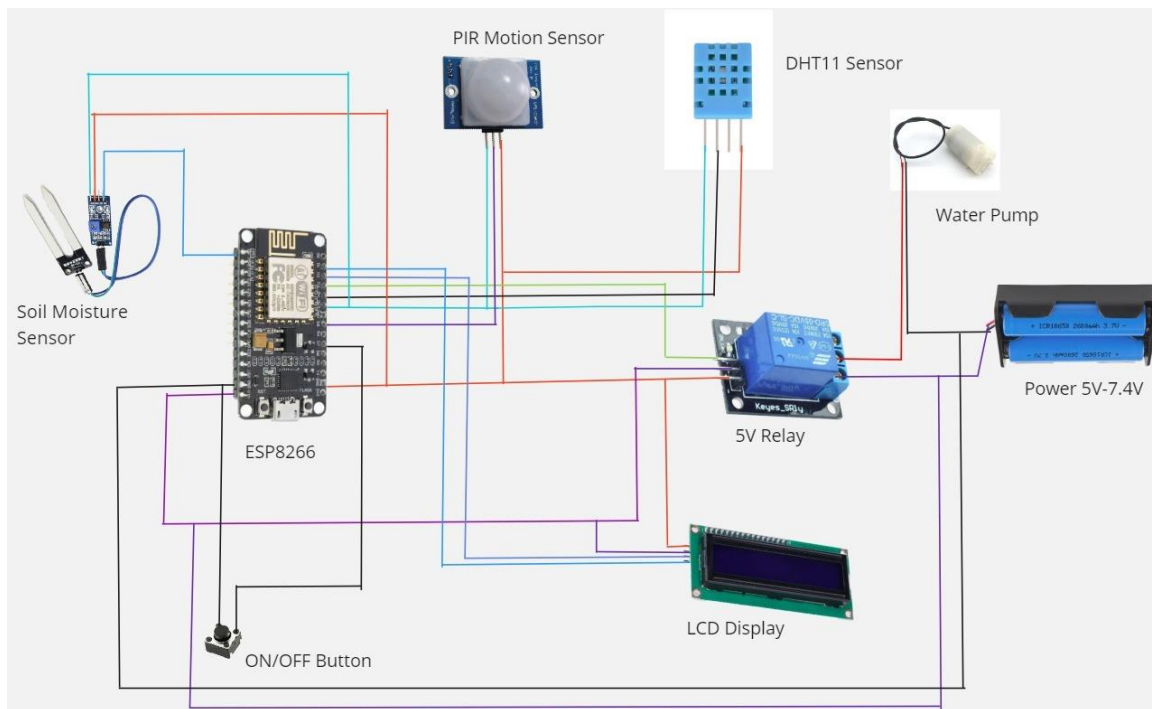
Objective:

- To make a successful automated system that ensures to keep the plants in the greenhouse healthy and hydrated with the nearest budget of Rs.5450 by 12th week.

SYSTEM DIAGRAM

Name of the item	Picture of the item
ESP8266	
PIR Motion Sensor	
DH11 Sensor	

Soil Moisture Sensor	
Water Pump Motor	
5V Relay	
LCD Display	
On and Off Button (Tactile Button)	
Breadboard	
Jumper Wires	



METHODOLOGY

Hardware components:

□ Microcontroller – ESP8266

A Wi-Fi enabled microcontroller that will act as the central processing unit, managing sensor inputs, controlling the water pump, and sending data to the cloud.

□ Sensor

- IR Motion Sensor - A motion sensor to detect the presence of insects, allowing the system to take preventive actions.
- DHT11 (or DHT22) - To provide real-time data on temperature and humidity making sure plant environment remains within optimal ranges for plant growth.
- Soil Moisture Sensor - Measures the moisture level on the soil and determine when to water the plants.

□ Actuators

- Water pump - To deliver water to the plants
- Relay Module – Based on sensor readings the rely module can control the power supply to various components, such as pumps and lights.

□ Power Supply

7.4 v battery is a compatible power source to the ESP8266 and the other components of the system.

□ Display

16x2 12C LCD Display to provide real-time information on temperature, humidity, soil moisture.

Additional components

- Breadboards
- Jump wires
- Resistors
- Transistors

Technologies:

□ Cloud services

For data storage and management, Real time monitoring and alerts and for backups and data security Cloud services can be useful in many ways.

□ Blynk Software

Blynk Software is a powerful platform that enables the creation of applications for controlling and managing Internet of Things (IoT) devices directly from your smartphone.

- Remote Monitoring and Control: Wherever you are, Blynk Software provides the ability to access and manage real-time data from various sensors, including temperature, humidity, and light conditions.
- Alerts and Notifications: Blynk Software delivers instant alerts, The system can notify you if motion sensors detect the presence of insects, enabling you to respond quickly to potential infestations.
- Automation and Scheduling: With Blynk Software, tasks like watering plants and controlling lighting can be automated and scheduled, minimizing the need for manual intervention.

EVALUATION METHOD

Plant Monitoring System

This evaluation can be done based on performance, reliability and affordability, usability and there are few steps to evaluate this system

1) Functionality testing

□ Water moisture sensor:

Test the testing accuracy of the soil moist level sensor, accurate data is received from mobile app & make sure that when watering mechanism through app should be activating if only it fulfils certain criteria where water levels are very low.

□ Heat sensor:

Probably the most crucial because the system must know that what its reading is in fact temperature, and whether it falls within a range suitable for plants if you want faster growth.

□ Motion sensor:

Here we are assessing; you can test that the motion sensitivity is high enough to detect a snake and not be tripped with fleeing pest as well making sure false positives will happen rarely.

2)usability testing

Testing for user friendliness in mobile App.

Installation challenges of the sensors using simple set up, pairing it with app and review how easy to comprehend data visibility & user instruction (usability).

3)performance testing

Check how long the sensors take to respond to an app.

This is a way we can see that performance.

Long term stability tests of the sensors.

4)reliability testing

Long term testing to validate the durability, accuracy, reliability of the sensor and systems.

Environmental: Different temperatures & humidity Specifications under which the system is available.

5) cost effectiveness testing

Determine the cost of your system.

How energy-efficient of the system is (to be determined after performing such testing).

BUDGET

Item	Cost (Rs).
ESP8266 Module	900/=
Soil Moisture Sensor	400/=
DHT11 Temperature and Humidity Sensor	300/=
16x2 I2C LCD Display	700/=
PIR Motion Sensor	300/=
Relay Module	250/=
Water Pump	600/=
7.4V Battery	1000/=
Miscellaneous Components (Includes jumper wires, breadboard, resistors, and other small components)	1000/=
Total Estimate	5450/=

★ *This is a rough estimate, and prices can vary based on the supplier and the quality of the items.*

REFERENCES

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